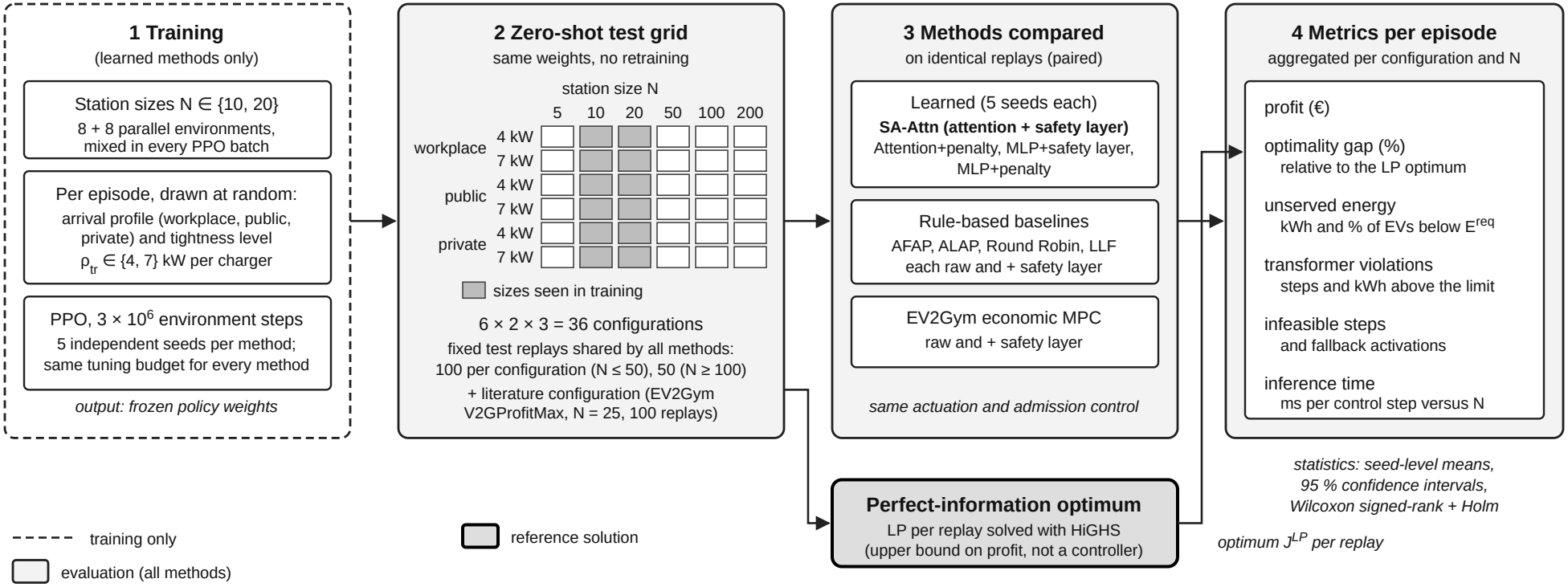
1 Training
(learned methods only)
Station sizes N ∈ {10, 20}
8 + 8 parallel environments, mixed in every PPO batch
Per episode, drawn at random:
arrival profile (workplace, public, private) and tightness level
ρ_{tr} ∈ {4, 7} kW per charger
PPO, 3 × 10^6 environment steps
5 independent seeds per method; same tuning budget for every method
output: frozen policy weights
2 Zero-shot test grid
same weights, no retraining
station size N
5 10 20 50 100 200
workplace 4 kW
7 kW
public 4 kW
7 kW
private 4 kW
7 kW
sizes seen in training
6 × 2 × 3 = 36 configurations
fixed test replays shared by all methods:
100 per configuration (N ≤ 50), 50 (N ≥ 100)
+ literature configuration (EV2Gym V2GProfitMax, N = 25, 100 replays)
3 Methods compared
on identical replays (paired)
Learned (5 seeds each)
SA-Attn (attention + safety layer)
Attention+penalty, MLP+safety layer, MLP+penalty
Rule-based baselines
AFAP, ALAP, Round Robin, LLF
each raw and + safety layer
EV2Gym economic MPC
raw and + safety layer
same actuation and admission control
Perfect-information optimum
LP per replay solved with HiGHS
(upper bound on profit, not a controller)
4 Metrics per episode
aggregated per configuration and N
profit (€)
optimality gap (%)
relative to the LP optimum
unserved energy
kWh and % of EVs below E^{req}
transformer violations
steps and kWh above the limit
infeasible steps
and fallback activations
inference time
ms per control step versus N
statistics: seed-level means, 95 % confidence intervals, Wilcoxon signed-rank + Holm
optimum J^{LP} per replay
training only
evaluation (all methods)
reference solution